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Class: CSE(AI)-SEDA Batch: B2

Assignment No.6 Use the diabetes data set from UCI and Pima Indians Diabetes data set for performing the following: a. Univariate analysis: Frequency, Mean, Median, Mode, Variance, Standard Deviation, Skewness and Kurtosis b. Bivariate analysis: Linear and logistic regression modeling c. Multiple Regression analysis d. Also compare the results of the above analysis for the two data sets.

Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.metrics import mean_squared_error, r2_score, accuracy_score
from scipy.stats import skew, kurtosis
from sklearn.metrics import confusion_matrix, precision_score, recall_score, f1_score, classification_report
```

Load Datasets

```
df_pima = pd.read_csv("diabetes.csv")
print("Pima Dataset Shape:", df_pima.shape)
df_pima.head()
```

```
Pima Dataset Shape: (768, 9)
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

Next steps: Generate code with df_pima New interactive sheet

```
diabetes = load_diabetes()
df_uci = pd.DataFrame(diabetes.data, columns=diabetes.feature_names)
df_uci['target'] = diabetes.target
print("UCI Dataset Shape:", df_uci.shape)
df_uci.head()
```

```
UCI Dataset Shape: (442, 11)
```

	age	sex	bmi	bp	s1	s2	s3	s4	s5	s6	target
0	0.038076	0.050680	0.061696	0.021872	-0.044223	-0.034821	-0.043401	-0.002592	0.019907	-0.017646	151.0
1	-0.001882	-0.044642	-0.051474	-0.026328	-0.008449	-0.019163	0.074412	-0.039493	-0.068332	-0.092204	75.0
2	0.085299	0.050680	0.044451	-0.005670	-0.045599	-0.034194	-0.032356	-0.002592	0.002861	-0.025930	141.0
3	-0.089063	-0.044642	-0.011595	-0.036656	0.012191	0.024991	-0.036038	0.034309	0.022688	-0.009362	206.0
4	0.005383	-0.044642	-0.036385	0.021872	0.003935	0.015596	0.008142	-0.002592	-0.031988	-0.046641	135.0

Next steps: Generate code with df_uci New interactive sheet

Data CLeaning

```
print("Pima Missing Values:\n", df_pima.isnull().sum())
print("UCI Missing Values:\n", df_uci.isnull().sum())
```

```
Pima Missing Values:
 Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64
UCI Missing Values:
 age       0
sex       0
bmi       0
bp        0
s1        0
s2        0
s3        0
s4        0
s5        0
s6        0
target    0
dtype: int64
```

Replace Invalid Zeros (Pima Dataset)

```
cols = ['Glucose','BloodPressure','SkinThickness','Insulin','BMI']
df_pima[cols] = df_pima[cols].replace(0, np.nan)

df_pima.fillna(df_pima.mean(), inplace=True)
df_pima.drop_duplicates(inplace=True)
```

Univariate Analysis

Frequency

```
print("Pima Outcome Frequency:\n", df_pima['Outcome'].value_counts())
print("\nUCI Target Frequency (Top 10):\n", df_uci['target'].value_counts().head(10))
```

```
Pima Outcome Frequency:
 Outcome
0    500
1    268
Name: count, dtype: int64

UCI Target Frequency (Top 10):
 target
200.0    6
72.0     6
178.0    5
71.0     5
90.0     5
97.0     4
55.0     4
53.0     4
131.0    4
150.0    4
Name: count, dtype: int64
```

Mean, Median, Mode

```
print("Pima Mean:\n", df_pima.mean())
print("UCI Mean:\n", df_uci.mean())

print("Pima Median:\n", df_pima.median())
print("UCI Median:\n", df_uci.median())

print("Pima Mode:\n", df_pima.mode().iloc[0])
print("UCI Mode:\n", df_uci.mode().iloc[0])
```

```
s3       -6.028360e-18
s4       -1.788100e-17
s5        9.243486e-17
s6        1.351770e-17
target    1.521335e+02
dtype: float64
Pima Median:
 Pregnancies                   3.000000
Glucose                     117.000000
BloodPressure                72.202592
SkinThickness                29.153420
Insulin                     155.548223
BMI                          32.400000
DiabetesPedigreeFunction      0.372500
Age                          29.000000
Outcome                       0.000000
dtype: float64
UCI Median:
 age         0.005383
sex        -0.044642
bmi        -0.007284
bp         -0.005670
s1         -0.004321
s2         -0.003819
s3         -0.006584
s4         -0.002592
s5         -0.001947
s6         -0.001078
target    140.500000
dtype: float64
Pima Mode:
 Pregnancies                   1.000000
Glucose                      99.000000
BloodPressure                70.000000
SkinThickness                29.153420
Insulin                     155.548223
BMI                          32.000000
DiabetesPedigreeFunction      0.254000
Age                          22.000000
Outcome                       0.000000
Name: 0, dtype: float64
UCI Mode:
 age        0.016281
sex       -0.044642
bmi       -0.030996
bp        -0.040099
s1        -0.037344
s2        -0.001001
s3        -0.013948
s4        -0.039493
s5        -0.018114
s6         0.003064
target    72.000000
Name: 0, dtype: float64
```

Variance & Standard Deviation

```
print("Pima Variance:\n", df_pima.var())
print("UCI Variance:\n", df_uci.var())

print("Pima Std Dev:\n", df_pima.std())
print("UCI Std Dev:\n", df_uci.std())
```

```
Pima Variance:
 Pregnancies                   11.354056
Glucose                      926.346983
BloodPressure                146.321591
SkinThickness                 77.280660
Insulin                     7228.588766
BMI                           47.267706
DiabetesPedigreeFunction       0.109779
Age                          138.303046
Outcome                        0.227483
dtype: float64
UCI Variance:
 age          0.002268
sex          0.002268
bmi          0.002268
bp           0.002268
s1           0.002268
s2           0.002268
s3           0.002268
s4           0.002268
s5           0.002268
s6           0.002268
target    5943.331348
```

```
dtype: float64
Pima Std Dev:
 Pregnancies                  3.369578
Glucose                     30.435949
BloodPressure               12.096346
SkinThickness                8.790942
Insulin                     85.021108
BMI                          6.875151
DiabetesPedigreeFunction     0.331329
Age                         11.760232
Outcome                      0.476951
dtype: float64
UCI Std Dev:
 age        0.047619
sex        0.047619
bmi        0.047619
bp         0.047619
s1         0.047619
s2         0.047619
s3         0.047619
s4         0.047619
s5         0.047619
s6         0.047619
target    77.093005
dtype: float64
```

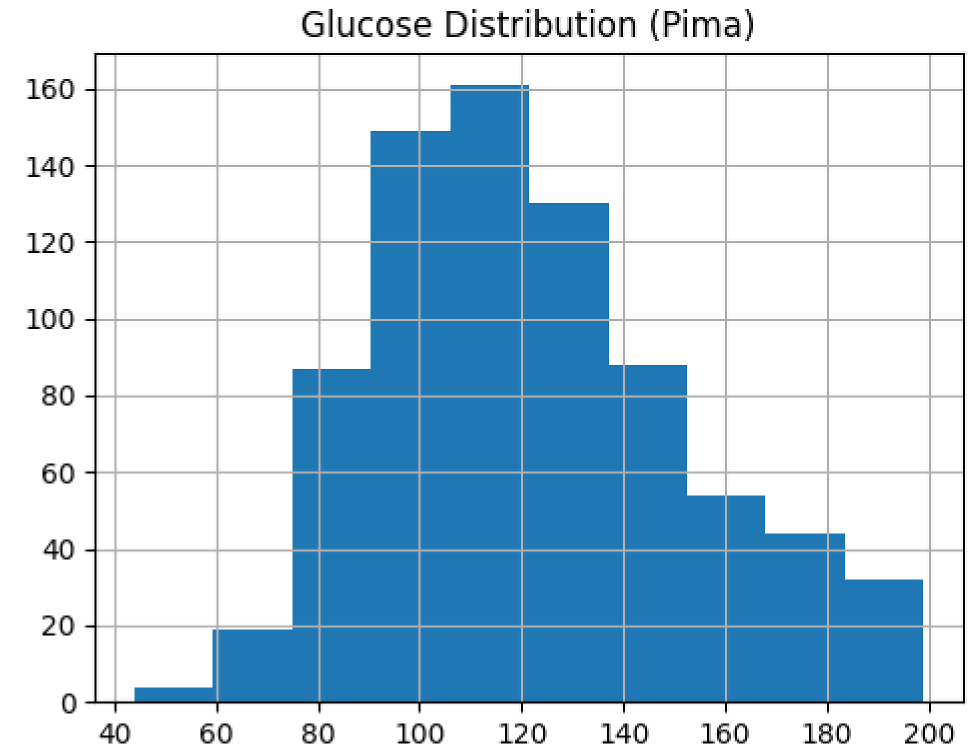
Skewness & Kurtosis

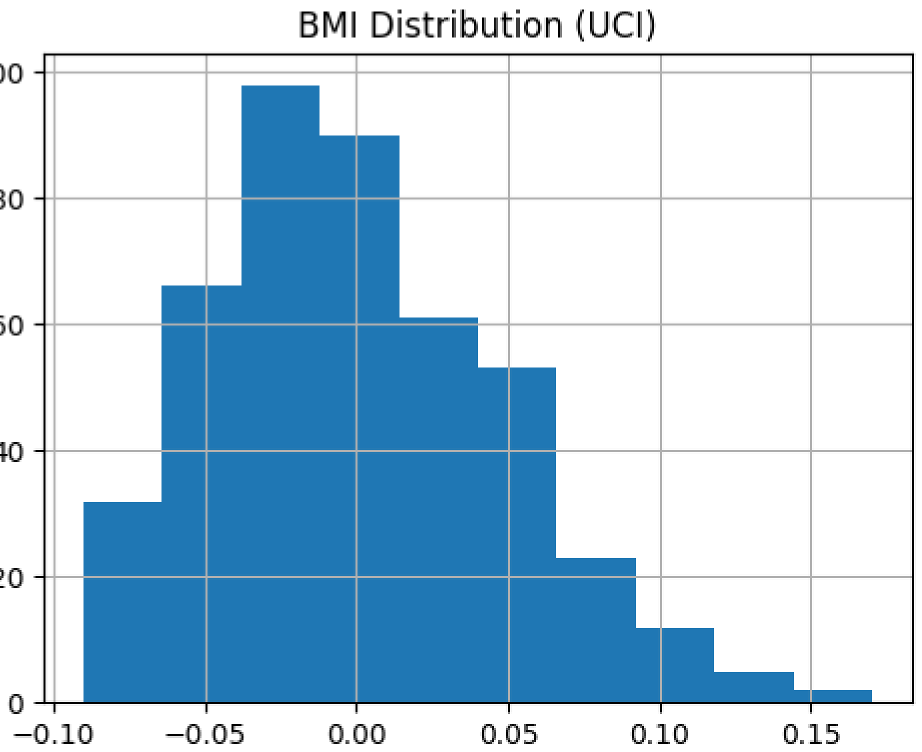
```
print("PIMA SKEWNESS & KURTOSIS")
for col in df_pima.columns:
    print(col, "Skew:", skew(df_pima[col]), "Kurtosis:", kurtosis(df_pima[col]))

print("\nUCI SKEWNESS & KURTOSIS")
for col in df_uci.columns:
    print(col, "Skew:", skew(df_uci[col]), "Kurtosis:", kurtosis(df_uci[col]))
```

```
PIMA SKEWNESS & KURTOSIS
Pregnancies Skew: 0.8999119408414357 Kurtosis: 0.15038273760948462
Glucose Skew: 0.531677628850459 Kurtosis: -0.2652767871408801
BloodPressure Skew: 0.1370370472689305 Kurtosis: 1.082847367848931
SkinThickness Skew: 0.8205664594675018 Kurtosis: 5.3718544792635345
Insulin Skew: 3.013183811523944 Kurtosis: 15.078748264774752
BMI Skew: 0.5970835583878779 Kurtosis: 0.9057125569396862
DiabetesPedigreeFunction Skew: 1.9161592037386292 Kurtosis: 5.550792047551203
Age Skew: 1.127389259531697 Kurtosis: 0.6311769413798585
Outcome Skew: 0.6337757030614577 Kurtosis: -1.5983283582089547

UCI SKEWNESS & KURTOSIS
age Skew: -0.23059556012398927 Kurtosis: -0.6771986900987508
sex Skew: 0.1269518188640044 Kurtosis: -1.983883235687121
bmi Skew: 0.5961166556214368 Kurtosis: 0.08047812866813064
bp Skew: 0.28967103827590446 Kurtosis: -0.5403332293804954
s1 Skew: 0.3768238227280956 Kurtosis: 0.2167770625244283
s2 Skew: 0.4351087583570202 Kurtosis: 0.5810556912433786
s3 Skew: 0.7965401530972942 Kurtosis: 0.9568955290670442
s4 Skew: 0.7328756796573327 Kurtosis: 0.42584638305986955
s5 Skew: 0.2907626799653419 Kurtosis: -0.14639566230288814
s6 Skew: 0.20721035056151146 Kurtosis: 0.22070114138631514
target Skew: 0.43906639932477265 Kurtosis: -0.8866436055681386
```

VISUALIZATION

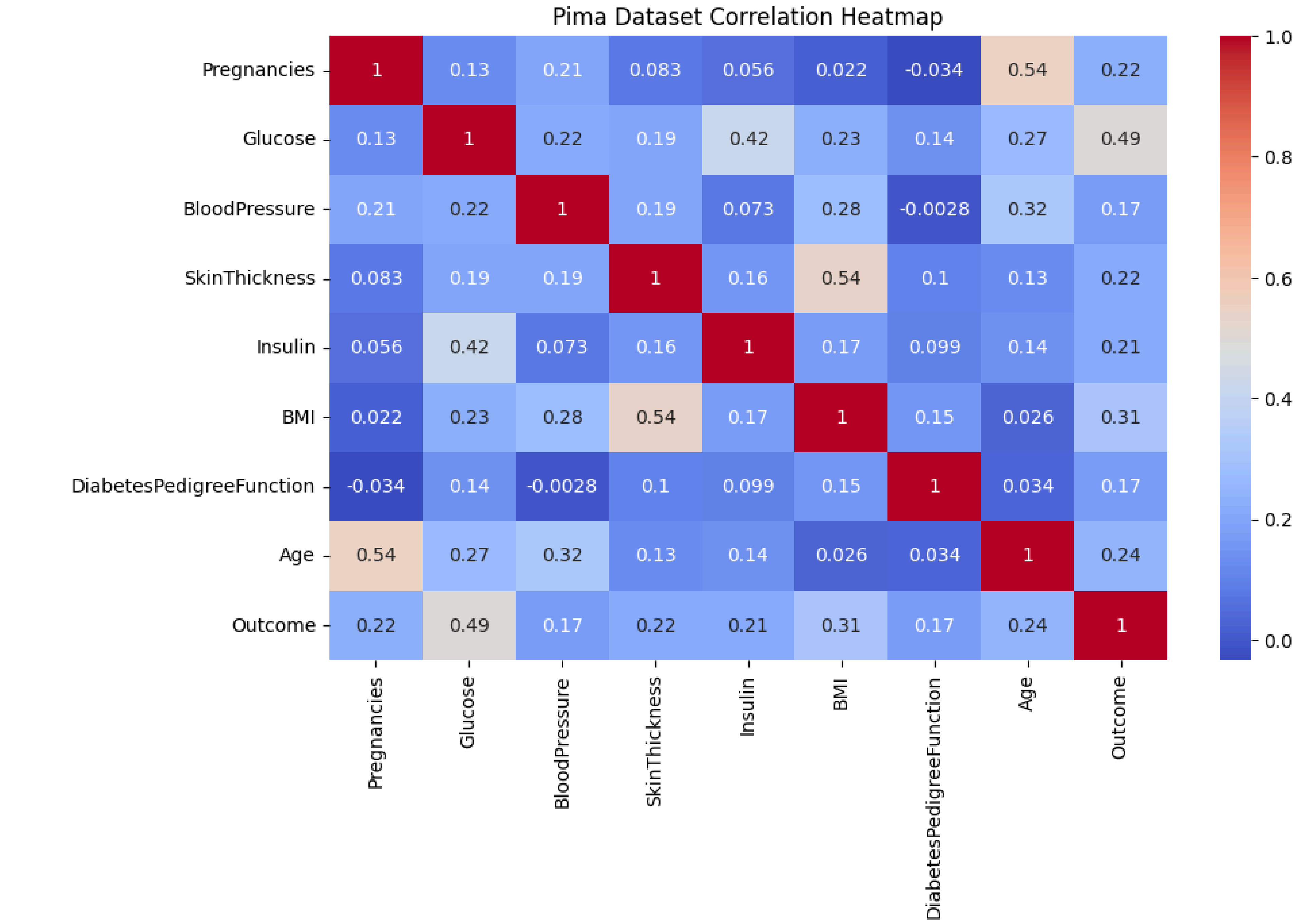
```
plt.figure(figsize=(5,4))
df_pima['Glucose'].hist()
plt.title("Glucose Distribution (Pima)")
plt.tight_layout()
plt.show()

plt.figure(figsize=(5,4))
df_uci['bmi'].hist()
plt.title("BMI Distribution (UCI)")
plt.tight_layout()
plt.show()
```

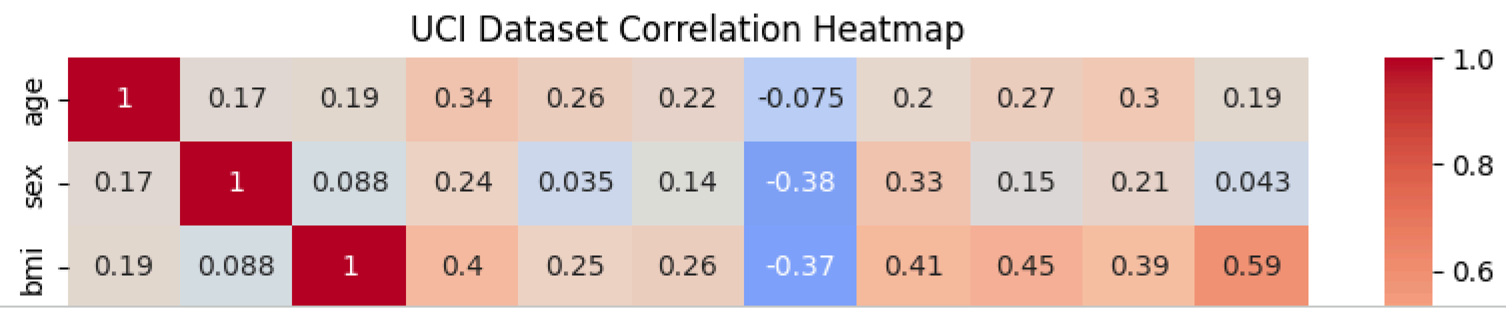




Correlation Heatmap

```
plt.figure(figsize=(10,6))
sns.heatmap(df_pima.corr(), annot=True, cmap='coolwarm')
plt.title("Pima Dataset Correlation Heatmap")
plt.show()

plt.figure(figsize=(10,6))
sns.heatmap(df_uci.corr(), annot=True, cmap='coolwarm')
plt.title("UCI Dataset Correlation Heatmap")
plt.show()
```





Bivariate Analysis

Linear Regression(UCI Dataset)

```
X_uci = df_uci.drop('target', axis=1)
y_uci = df_uci['target']

X_train_u, X_test_u, y_train_u, y_test_u = train_test_split(
    X_uci, y_uci, test_size=0.2, random_state=42)

lin_model = LinearRegression()
lin_model.fit(X_train_u, y_train_u)

y_pred_u = lin_model.predict(X_test_u)

print("UCI Linear Regression")
print("MSE:", mean_squared_error(y_test_u, y_pred_u))
print("R2 Score:", r2_score(y_test_u, y_pred_u))
```

```
UCI Linear Regression
MSE: 2900.193628493482
R2 Score: 0.4526027629719195
```

Logistic Regression (Pima Dataset)

```
X_pima = df_pima.drop('Outcome', axis=1)
y_pima = df_pima['Outcome']

X_train_p, X_test_p, y_train_p, y_test_p = train_test_split(
    X_pima, y_pima, test_size=0.2, random_state=42)
```

```
log_model = LogisticRegression(max_iter=2000)
log_model.fit(X_train_p, y_train_p)

y_pred_p = log_model.predict(X_test_p)

print("Pima Logistic Regression")
print("Accuracy:", accuracy_score(y_test_p, y_pred_p))
```

```
Pima Logistic Regression
Accuracy: 0.7532467532467533
```

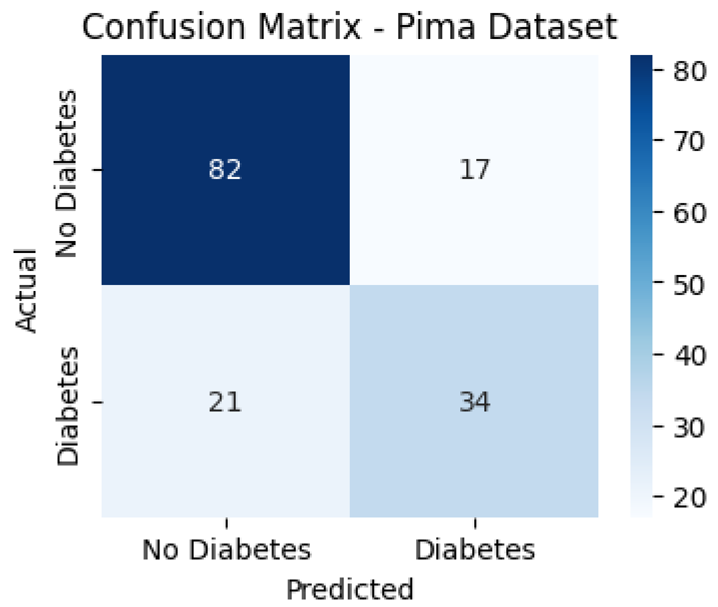
MULTIPLE REGRESSION ANALYSIS(UCI)

```
multi_model = LinearRegression()
multi_model.fit(X_train_u, y_train_u)

print("Multiple Regression R2 Score:",
      multi_model.score(X_test_u, y_test_u))
```

```
Multiple Regression R2 Score: 0.4526027629719195
```

Confusion Matrix

```
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test_p, y_pred_p)

plt.figure(figsize=(4,3))
sns.heatmap(cm,
            annot=True,
            fmt='d',
            cmap='Blues',
            xticklabels=['No Diabetes','Diabetes'],
            yticklabels=['No Diabetes','Diabetes'])

plt.title("Confusion Matrix - Pima Dataset")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



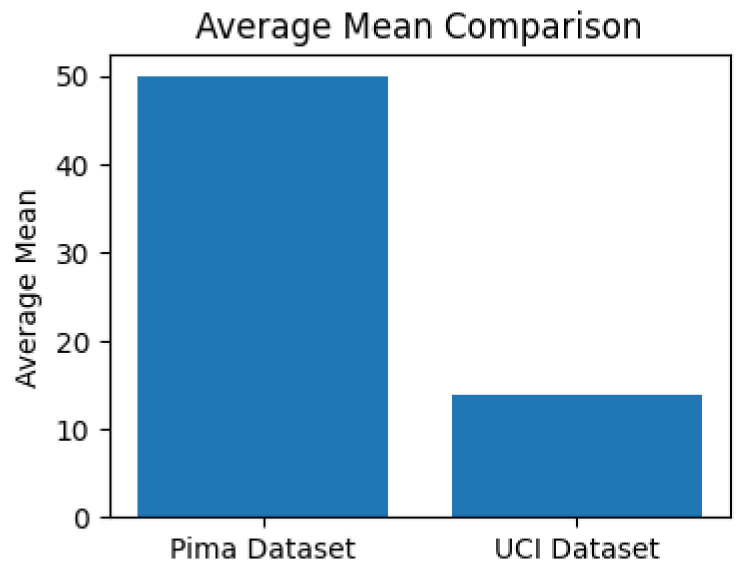
Precision, Recall, F1

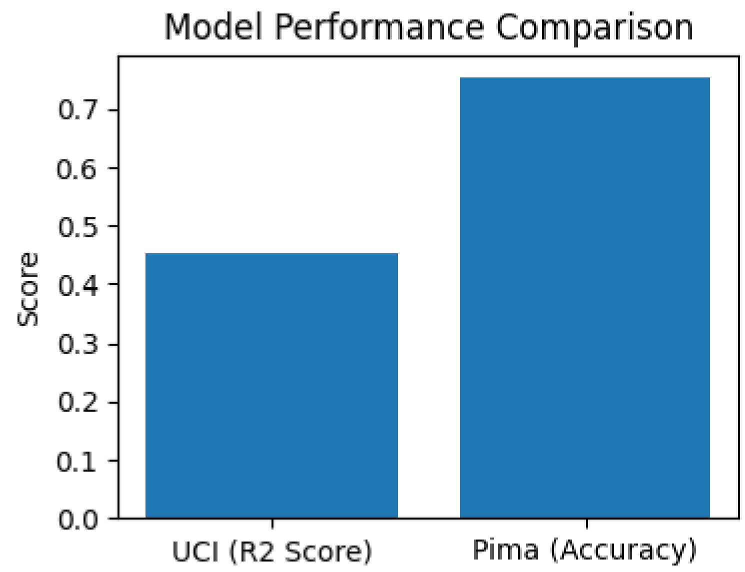
```
from sklearn.metrics import precision_score, recall_score, f1_score, classification_report

print("Precision:", precision_score(y_test_p, y_pred_p))
print("Recall:", recall_score(y_test_p, y_pred_p))
print("F1 Score:", f1_score(y_test_p, y_pred_p))

print("\nClassification Report:\n",
      classification_report(y_test_p, y_pred_p))
```

```
Precision: 0.6666666666666666
Recall: 0.6181818181818182
F1 Score: 0.6415094339622641

Classification Report:
               precision    recall  f1-score   support
```

```
           0       0.80      0.83      0.81        99
           1       0.67      0.62      0.64        55

    accuracy                           0.75       154
   macro avg       0.73      0.72      0.73       154
weighted avg       0.75      0.75      0.75       154
```

```
pima_mean = df_pima.mean().mean()
uci_mean = df_uci.mean().mean()

plt.figure(figsize=(4,3))
plt.bar(['Pima Dataset','UCI Dataset'],
        [pima_mean, uci_mean])
plt.title("Average Mean Comparison")
plt.ylabel("Average Mean")
plt.show()

uci_r2 = r2_score(y_test_u, y_pred_u)
pima_accuracy = accuracy_score(y_test_p, y_pred_p)

plt.figure(figsize=(4,3))
plt.bar(['UCI (R2 Score)','Pima (Accuracy)'],
        [uci_r2, pima_accuracy])
plt.title("Model Performance Comparison")
plt.ylabel("Score")
plt.show()
```






```
plt.figure(figsize=(5,4))
plt.bar(['Pima','UCI'], [pima_mean, uci_mean])
plt.title("Mean Comparison")
plt.tight_layout()
plt.show()
```